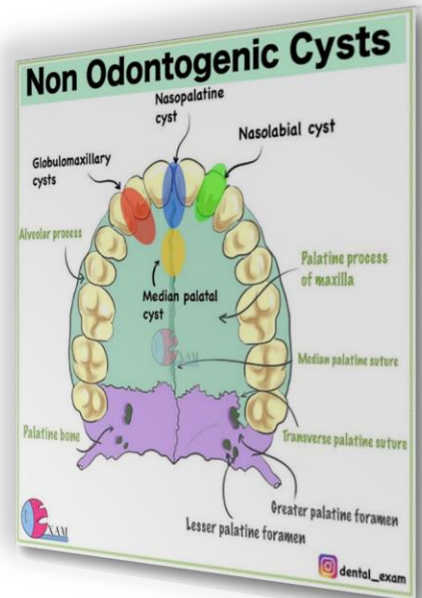
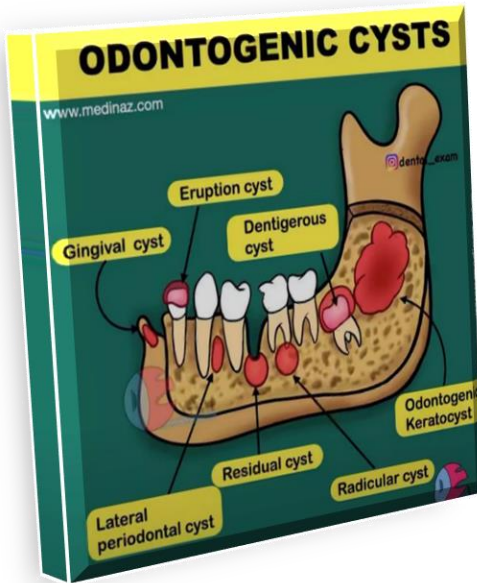




Cysts of the JAW (2026)



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CYSTS OF THE ORAL AND MAXILLOFACIAL REGION



Classify and identify cysts of the jaws and soft tissue

Understand the etiology and pathogenesis of jaw cysts

Recognize the clinical presentation of jaw cysts

Interpret radiographic findings

Understand the principles of treatment

Manage complications and recurrence

CYSTS OF THE ORAL AND MAXILLOFACIAL REGION

Definition of cyst

The cyst is defined as a pathological cavity filled with fluid or semi-fluid, or gaseous contents and not created by the accumulation of pus". It may have an epithelial lining called (**a true cyst**) OR a lack of epithelial lining called (**a pseudo-cyst**).

Complication of cysts

- o **Infection:** Cyst is liable to be infected can cause osteomyelitis, facial cellulitis, etc.
- o **Pathological fracture** of bone.
- o **Disfigurement** of face.
- o **Neoplasm** may arise from the epithelial lining.
- o Some cyst e.g. dentigerous cysts may **prevent the eruption** of the involved tooth.

Parts of cyst

generally, the cyst is formed from:

1. Epithelial lining:

Most of the cysts are lined wholly or in part by epithelial lining. This lining can be a stratified squamous epithelium, pseudostratified epithelium, or simple epithelium.

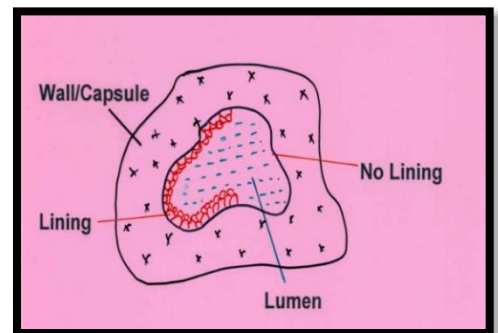


FIGURE 1: SCHEMATIC OF TYPICAL CYST

2. Cystic cavity or lumen: the cystic cavity, generally consists of:

- Water and electrolytes, Breakdown products of degenerating epithelial, inflammatory, connective tissue cells and Serum proteins.

3. Fibrous tissue capsule: which is formed from connective tissue cells such as fibroblasts and fibrocytes as well as collagen fibers.

Pathogenesis of cyst

The formation of a cyst takes place in three phases:

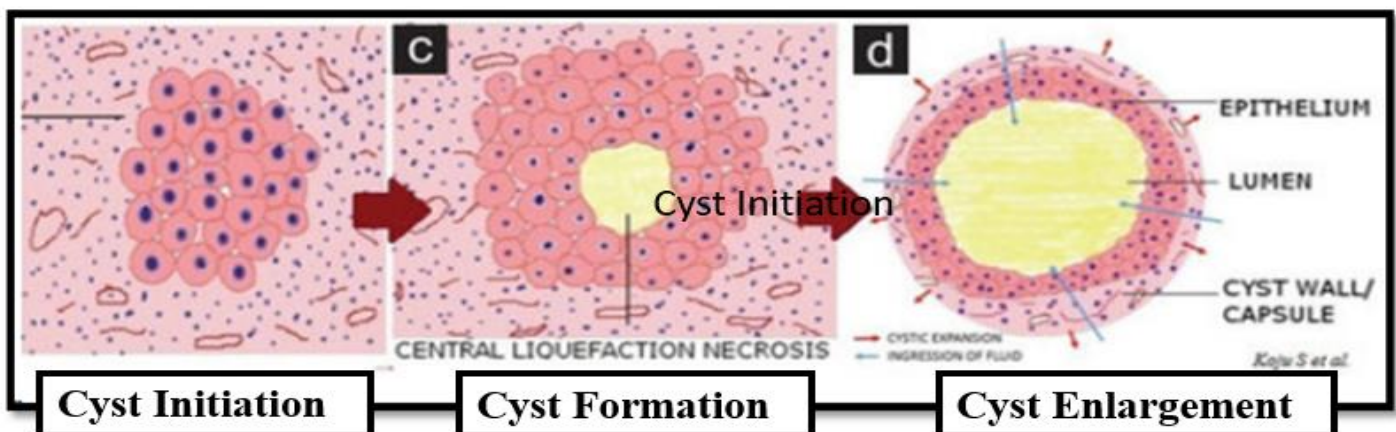
- I. **Initiation:** the factors depend on the type of cyst
- II. **Cyst formation**
- III. **Cyst expansion** or enlargement

I. Cyst Initiation: An unknown stimulus promotes the rapid increase of epithelial rest cells via cytokine synthesis. The factors that are suggested to be responsible for the cyst initiation phase are mainly genetics, loss of immunological surveillance, inflammatory mediators, or some local factors like decreased oxygen tension along with increased CO₂ tension. Initiation is followed by the rapid growth of the epithelial cells and the development of a cystic lesion.

II. Cyst Formation: Nutritional deficiency theory

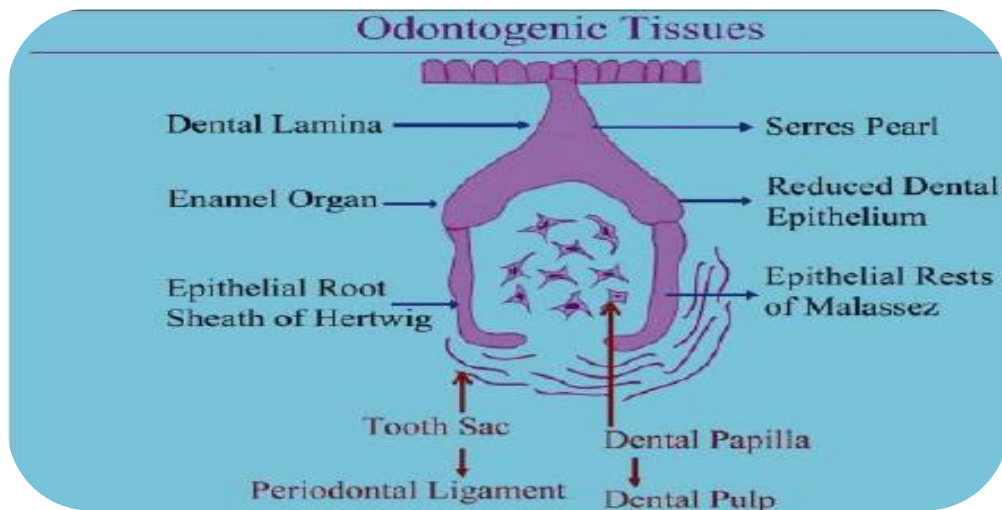
After initiation, proliferating epithelial cells form a mass inside the sac, and the innermost central cells become away from nutrients/blood supply, as they are far from the source of nutrients. The innermost cells do not get adequate blood supply, and so there is ischemic *liquefactive necrosis* in the center, leading to cavity formation, which is surrounded by growing epithelial cells.

III. Cyst Enlargement or Expansion: Once formed, the cyst continues to enlarge slowly, over the months. The process is similar for all epithelial-lined cysts with some variations. The basic mechanism for enlargement is the same for most cysts.



The Origin of Epithelial Lining of Cyst

- ❖ **The epithelial lining of (Odontogenic Cysts)** ➡ originates from residues of the tooth-forming organ. *There are three types:*
 1. **Epithelial rests of Malassez:** Formed from the fragmentation of the epithelial root sheath of Hertwig.
 2. **The epithelial rests of Serres:** Persisting after the dissolution of the dental lamina.
 3. **The reduced enamel epithelium:** Derived from an enamel organ and covers the fully formed crown of an unerupted tooth.
- ❖ **The epithelial lining of (Non-odontogenic Cysts)** ➡ originates from epithelial cells entrapped between developing facial processes or the epithelial lining of a fistulous tract after abscess healing.



Type	Source	Origin of cell rest	Cyst variety
Odontogenic rests	Cell rests of malassez	Epithelial hertwig's root sheath	Periapical (radicular) cyst
	Reduced enamel epithelium	Enamel organ	Dentigerous cyst
	Rests of dental lamina (rest of Serres)	Epithelial connection between mucosa and enamel organ	Glandular odontogenic cyst, Lateral periodontal cyst, Odontogenic keratocyst, and Gingival cyst of newborn & adult
Nonodontogenic Rests	Remnants of nasopalatine duct	Paired nasopalatine duct-vestigial	Nasopalatine canal cyst and Fissural cysts
	Remnants of maxillary sinus epithelium		Cysts of the maxillary sinus

WHO (2022) Classification of cysts of the jaws

A. Epithelialized cysts

1) Odontogenic cysts

a) Inflammatory Cysts

1. Radicular (Periapical) cyst
 - i. Residual Cyst
 - ii. Lateral radicular Cyst
2. Inflammatory Collateral Cysts:
 - i. Mandibular buccal bifurcation cyst
 - ii. Paradental cyst

b) Development Cysts

1. Dentigerous cyst
2. Eruption cyst
3. Odontogenic Keratocyst (OKC)
4. Orthokeratinized odontogenic cyst
5. Lateral periodontal cyst and botryoid odontogenic cyst
6. Gingival cyst of the adult
7. Gingival (alveolar) cyst of the newborn
8. Glandular odontogenic cyst
9. Calcifying odontogenic cyst (COC)

2) Non-odontogenic cysts

1. Post surgical ciliated cyst
2. Nasopalatine duct (incisive canal) cyst
3. Nasolabial cyst
4. Median palatal cyst

B. Non-epithelialized primary bone cysts

1. Solitary bone cyst
2. Aneurysmal bone cyst
3. Stafine's idiopathic bone cavity

N.B: Cysts of Soft tissues

1. Dermoid & epidermoid cyst
2. Lymphoepithelial cyst
3. Thyroglossal duct cyst
4. Salivary gland cysts

The General Features of Cyst

1) **Origin:** in the case of true cysts:

- **Odontogenic cysts:** arising from epithelial remnants of the odontogenic epithelium (e.g. epithelial rests of Serres, reduced enamel epithelium, or epithelial rests of Malassaz).
- **Non-odontogenic cysts:** epithelial remnants other than tooth-forming cells.

2) **Clinically:**

- **Age:** second and third decades (middle age) **Except:**
 - ✓ Inflammatory radicular cysts occur **at any age** but **more in adults**.
 - ✓ Eruption cysts occur in **children**.
 - ✓ Gingival cysts of newborns occur in **newborns**.
 - ✓ Gingival cysts of adults occur in adults **over 40 years**.
- **Site:** Mandible more than Maxilla.
 - ✓ When present in the mandible, it occurs more posteriorly.
 - ✓ When they occur in the maxilla, they present more in the anterior region.
- **Symptoms:**
 - ❖ **In case of jaw cysts:**
 - ✓ Cysts are **symptomless** and discovered by routine x-ray.
 - ✓ Cysts are **painless** unless **infected** (**except** aneurysmal bone cyst which is **painful** and tender upon motion).
 - ✓ Cysts are slowly growing.
 - ❖ **In case of soft tissue cysts:**
 - ✓ Painless and slowly growing.
 - ✓ Soft, fluctuant, and dome shape.
 - ✓ It may be doughy in consistency if it contains keratin.
 - ✓ Its color may be normal (e.g. retention cyst), bluish (e.g. extravasated mucocele, ranula, and eruption cyst), or white (e.g. gingival cyst of the newborn due to the presence of keratin).
 - ✓ The cysts may be covered with normal skin (e.g. lymphoepithelial and thyroglossal tract cysts).

3) **Histopathological:**

- ❖ **Epithelial lining:**
 - ✓ In most cases it is stratified squamous epithelium (keratinized or nonkeratinized).
 - ✓ Respiratory epithelium (pseudo-stratified ciliated columnar epithelium).
- ❖ **Fibrous connective tissue wall:** it consists of fibroblasts and collagen fibers.

Inflammatory Odontogenic Cysts

1-PERIAPICAL CYST (RADICULAR CYST, APICAL PERIODONTAL CYST)

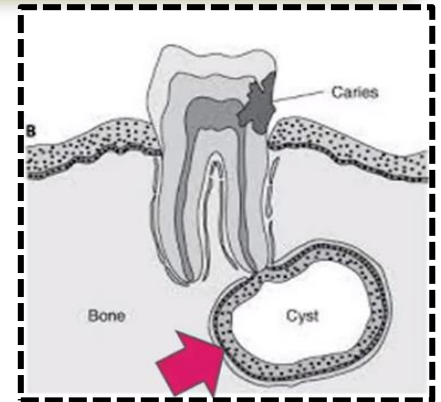
❖ Definition:

- It is *the commonest cystic lesion* of the jaws and (the most common odontogenic cyst of the oral cavity).
- It is **an inflammatory** odontogenic cyst that occurs at **the periapical area of non-vital tooth**.

- ❖ Origin: Epithelial lining derived from the proliferation of **epithelial rests of Malassez (ERM)** within the periodontal ligament at the apex of the tooth.

❖ Etiology & Pathogenesis:

- A periapical cyst develops from *a preexisting periapical granuloma*, which is chronically inflamed granulation tissue located at *the apex of a nonvital tooth*.



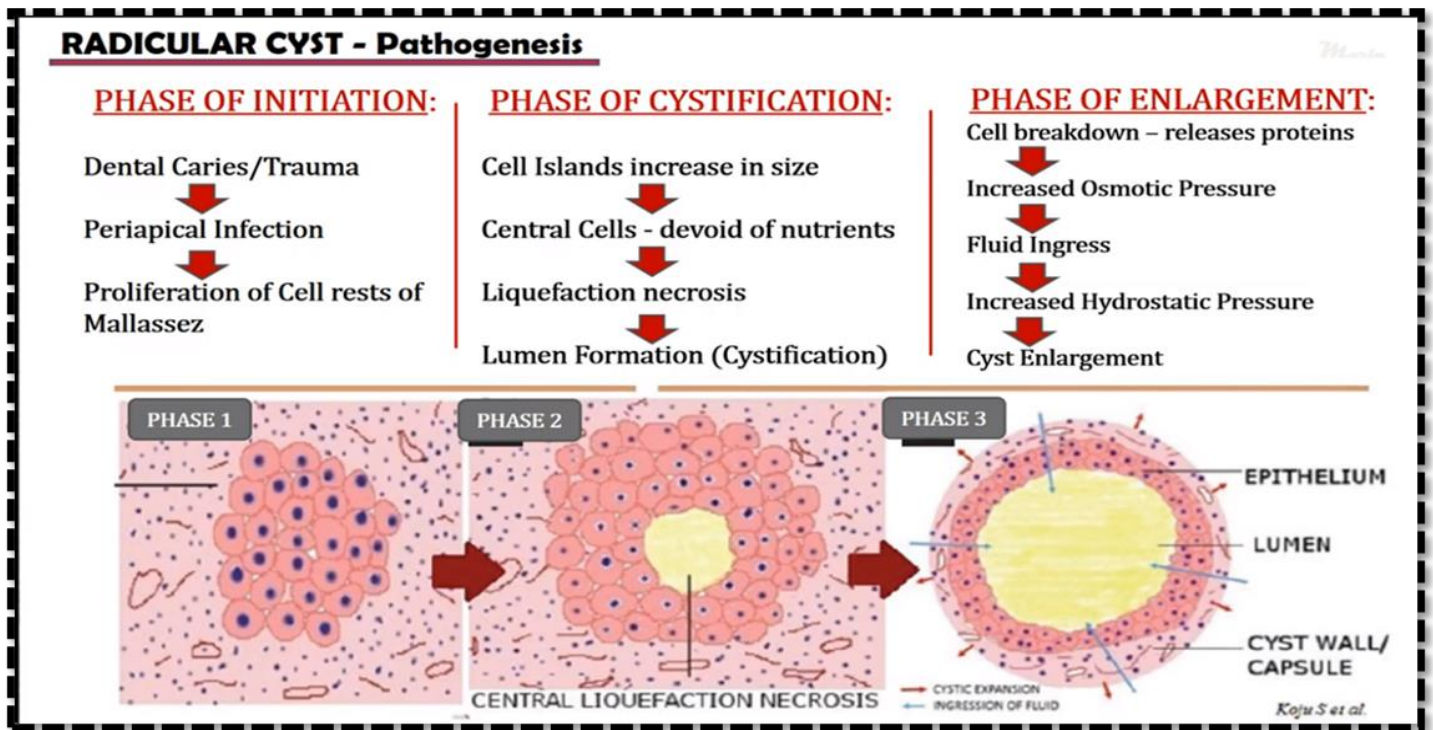
Pathogenesis of Radicular Cysts:

- ✓ Radicular cyst arise from undergoes proliferation of the epithelial rests of malassez in response to the products of inflammation which lead to form of masses.
- ✓ As theses masses enlarge, their central cells become away from the source of nutrition and they degenerated and forming microcyst which enlarge and finely fuse together forming one large macrocyst.

Mechanism of Cyst Enlargement:

- Breakdown of the complex tissue proteins that derived from degenerating epithelial and connective tissue cells into simpler protein resulted in an increase in the osmotic pressure inside the cyst.
- The epithelial lining acts as a semi-permeable membrane through which the fluid is transported from outside to the cystic cavity, which leads to an increase in the osmotic pressure within the cyst.

- In addition, cyst lining and fibroblasts in the cyst capsule release bone-resorbing factors such as prostaglandins, especially PGE2 and PGF3 and interleukin-1, which can assist in the process of bone resorption by creating more space for cyst enlargement.

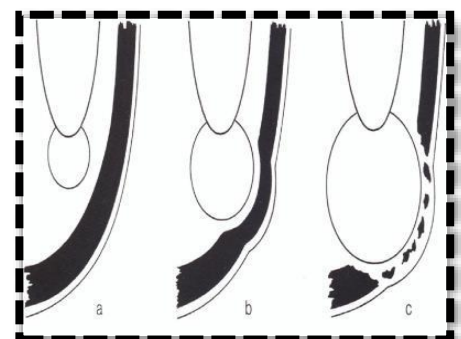


Clinical Features:

- Incidence & Frequency:** Most common cyst of the jaw, accounting for (60-75%) of all jaw cysts.
- Age:** Peaks in the 3th and 6th decades. Rarely occurring in the first decade of life.
- Sex:** Slightly more in **males**.
- Site predilection:** Most commonly occurs in **maxillary anterior permanent teeth**.
- Signs & symptoms:**
 - ✓ Painless (asymptomatic) discovered accidentally during routine radiographic examination which is *always associated with the non-vital tooth*.
 - ✓ Slowly enlarging painless swelling on labial or buccal sides.

The swelling usually runs across 3 stages

- Initially**, it slowly expands the cortical plate at a rate of about 5mm in diameter per year and produces a bony-hard expansion.



- b. **Later**, when the rate of expansion increases, this leads to progressive thinning of the cortical plate to produce a clinical sign of (**egg-shell crackling**).
- c. **Finally**, the cyst perforates the cortical plate and presents clinically as a bluish, fluctuant sub-mucosal swelling.

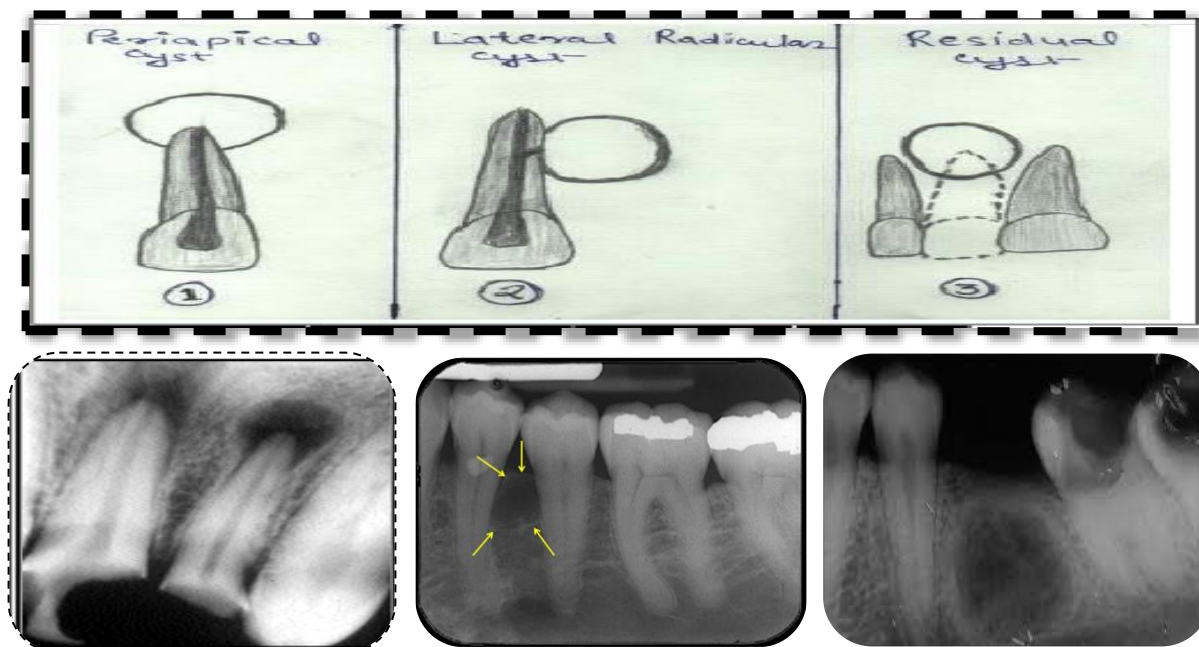
Radiographic Features:

- Round or ovoid well-defined radiolucency area surrounded by a narrow radiopaque margin around the apex of a non-vital tooth.
- The size of the cyst is about 5mm in diameter.
- The opaque margin becomes hazy if the cyst is secondarily infected.
- In long-standing cysts, root resorption of the associated tooth may occur.

Types of radicular cyst radiographically

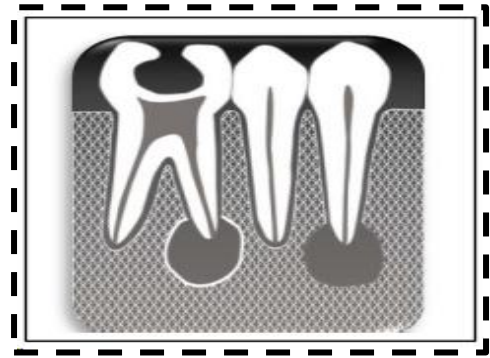
Radicular cysts are subdivided into three types depending on the location of the cyst:

1. **Apical radicular cyst:** most common type of radicular cyst (about 70%). It arises as a result of the extension of inflammation from the pulp to the periapical part of the tooth (around the apex of the root).
2. **Lateral radicular cyst:** appears along the lateral aspect of the root, as the inflammation spreads through a lateral foramen. Radiographically, it appears as discrete radiolucencies along the lateral aspect of the root.
3. **Residual periapical cyst:** If periapical inflammatory tissue is not curetted at the time of tooth removal, it may give rise to an inflammatory cyst called a residual inflammatory cyst. It appears as a round-to-oval radiolucency of variable size within the alveolar ridge at the site of the previous tooth extraction.



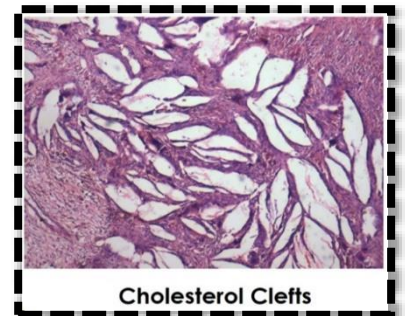
Differential Diagnosis of Periapical Radiolucency:

1. Periapical granuloma.
2. Radicular cyst.
3. Chronic periapical abscess.
4. Periapical fibrous scar.
5. Periapical cemento-osseous dysplasia (early stage)



Histopathological Features:

- Cystic cavity is lined by *non-keratinized stratified squamous epithelium* of variable thickness.
- In newly formed cysts, the epithelial lining is hyperplastic, exhibiting long anastomosing cords of epithelium forming arcades or rings over vascularized CT and many inflammatory cells.
- In fully formed cysts (old cysts) the epithelial lining becomes more regular and flattened, and the CT is more fibrous & less vascular, with fewer inflammatory cells.
- **Connective tissue contains:**
 - ✓ Variable number of inflammatory cells; plasma cells are also present and may be degenerated (**Russell bodies**).
 - ✓ Multinucleated foreign body giant cells MNGC are seen especially in areas of cholesterol clefts.
 - ✓ **Cholesterol clefts** are seen in CT and may erode through the epithelium and protrude into the cyst lumen (mural nodule).
 - ✓ **Foam cells** (lipid-filled macrophages).
- **Cyst contents:**
 - ✓ The cyst contents vary from a watery, straw-colored fluid to a semi-solid brownish paste-like material. The cyst consists of all serum proteins. The soluble protein level is 5-11g/dl.
 - ✓ Water, electrolytes, cholesterol crystals, and breakdown products of degenerating, necrotic epithelial, connective tissue, and inflammatory cells.



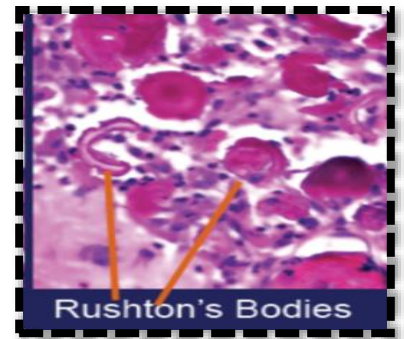
Microscopic Variation in Cyst Lining & Wall of Radicular Cyst

1. **Metaplasia** of the epithelial lining to the respiratory epithelium (pseudostratified columnar ciliated cells with goblet cells).
2. **Foci of dystrophic calcification:** Basophilic (haematoxyphilic) calcific deposits may be found in both the epithelium and CT.

3. **Hyaline bodies or “Rushton bodies”:** These are thin, Eosinophilic (acidophilic), linear, hairpin, or slightly curved bodies with concentric lamination. If present, they are only seen in the epithelium.

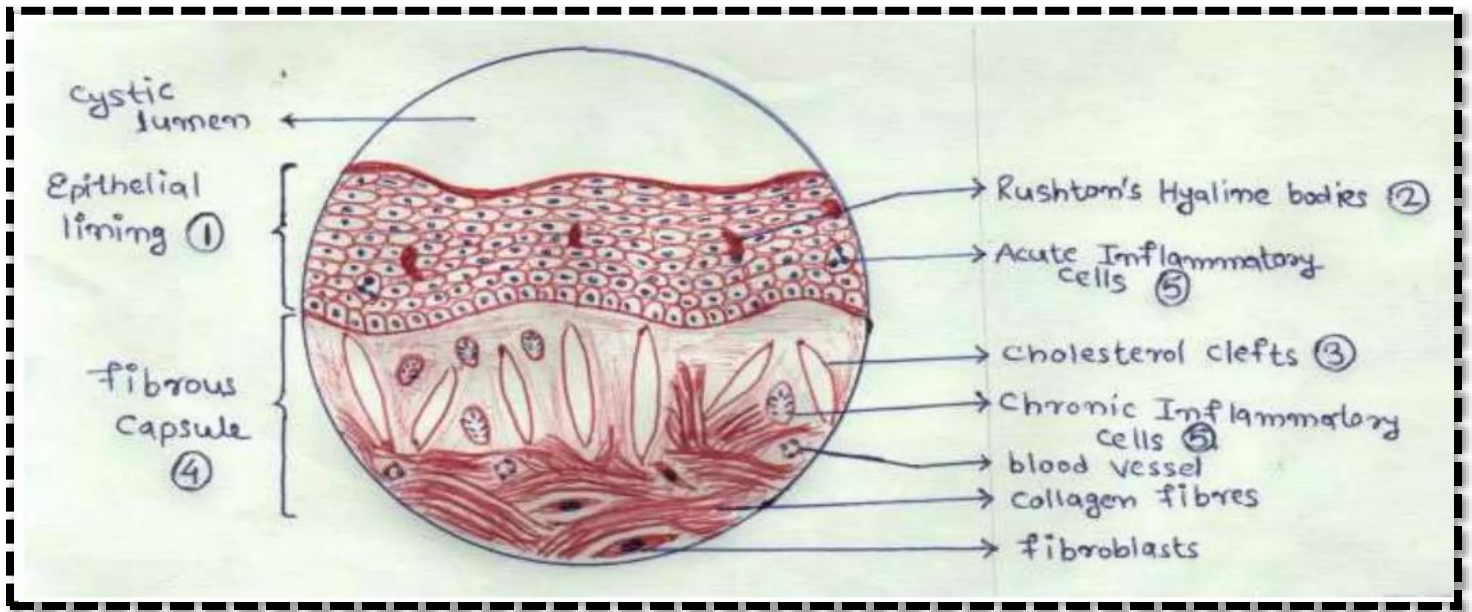
➤ **The origin of Hyaline bodies:**

- Haematogenous origin related to previous hemorrhage.
- Odontogenic origin.



Treatment and Prognosis

- Root Canal Treatment (RCT) + Apicoectomy + Enucleation.
- Extraction of the involved tooth & curettage of the apical zone.



2-Inflammatory Collateral Cyst

- It's an **uncommon inflammatory odontogenic cyst** that arises on the disto-buccal aspect of the roots of partially or recently erupted teeth.
- They arise as a result of **inflammation in the pericoronal tissues**; it may induce periodontal pocketing, which may lead to hyperplasia and cyst formation in adjacent odontogenic epithelium, which may be either remnants of the reduced enamel epithelium or cell rests of Malassez.
- The cysts were lined with a non-keratinised squamous epithelium with a mixed inflammatory cell infiltrate in the connective tissue

➤ **Two main types of inflammatory collateral cysts are recognized:**

- i-** Mandibular infected buccal cyst (Buccal bifurcation Cyst)
- ii-** Paradental cyst

i- Buccal Bifurcation Cyst

Definition:

It is an uncommon inflammatory odontogenic cyst that characteristically develops on the a buccal aspect of **the mandibular first permanent molar**.

Etiology & Pathogenesis:

The pathogenesis of this cyst is *uncertain*, it has been speculated that when the tooth erupts, an inflammatory response may occur in the surrounding follicular tissues that stimulates cyst formation.

Clinical Features:

- It typically occurs in **children from 5 to 11 years of age**.
- The patient has slight to moderate tenderness on the buccal aspect of the mandibular first molar, which may be *in the process of erupting*.
- The patient often notes associated clinical swelling and a foul-tasting discharge.
- Periodontal probing usually reveals pocket formation on the buccal aspect of the involved tooth.

Radiographic features:

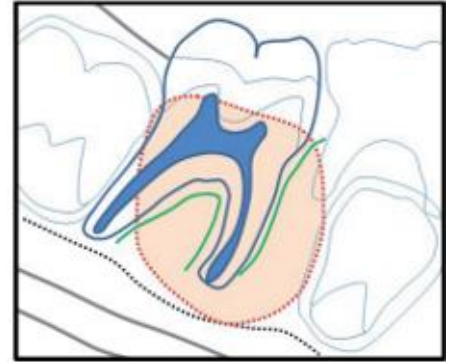
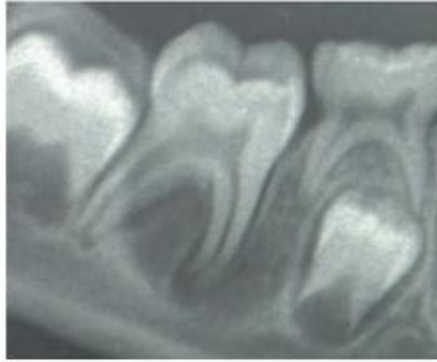
- Typically show a **well-circumscribed unilocular radiolucency involving the buccal furcation and root area of the involved tooth**.
- An *occlusal radiograph* is most helpful in demonstrating the buccal location of the lesion (The root apices of the molar are characteristically tipped toward the lingual mandibular cortex).

Histopathologic Features

- The microscopic features are nonspecific and show a cyst that is lined by nonkeratinizing stratified squamous epithelium with areas of hyperplasia.
- A prominent chronic inflammatory cell infiltrate is present in the surrounding connective tissue wall.

Treatment and Prognosis

The buccal bifurcation cyst is usually treated by enucleation; extraction of the associated tooth is unnecessary.



ii-Paradental Cyst

Definition:

- The term paradental cyst has sometimes been used synonymously for buccal bifurcation cyst.
- Such lesions typically occur distal or buccal to partially erupted mandibular third molars with a history of pericoronitis.



Etiology & Pathogenesis:

- The pathogenesis of the paradental cyst is also uncertain. Some of these lesions have been associated with teeth that demonstrate buccal enamel extensions into the bifurcation area.
- Such extensions may predispose these teeth to buccal pocket formation, which could then enlarge to form a cyst in response to pericoronitis.
-

Clinical features:

Paradental cyst arises alongside a partially erupted third molar involved by pericoronitis

Radiographic features:

Appears as a well-defined area of radiolucency related to the neck of a partially erupted tooth and the coronal third of the root.



Histopathological features:

A paradental cyst is similar to an inflammatory radicular cyst.

Developmental Odontogenic Cysts

1-DENTIGROUS CYST (Follicular Cyst)

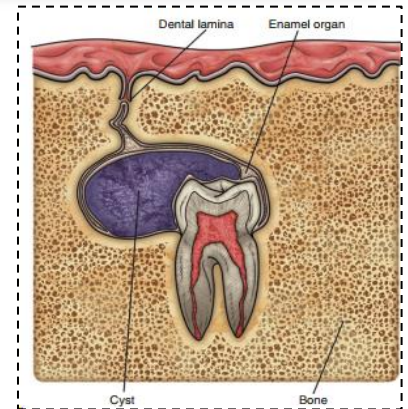
❖ Definition:

- ✓ Is the developmental odontogenic cyst which encloses **a part or entire crown of an unerupted tooth** and is attached to the tooth at **the cemento-enamel junction**.

✚ **N.B:** Dentigerous cysts, or follicular cysts: the term dentigerous is a Latin word that literally means “tooth-bearing/producing”

❖ Origin:

Enamel organ or remnant "Reduced enamel epithelium".



❖ Etiology & Pathogenesis:

The pathogenesis of this cyst is **uncertain**, but apparently, it develops by an accumulation of fluid between the *reduced enamel epithelium* and the tooth crown & the separation of the follicle from around the crown of an unerupted tooth.

❖ Clinical Features

- **Incidence and Frequency:** Dentigerous cyst is **the second** most common type of odontogenic cyst and **the most common** developmental cyst of the jaws (20% of all epithelium-lined cysts of the jaws).
- **Age:** Most frequently present in patients between **(10 and 30 years)** of age.
- **Sex:** There is a slight **male** predilection.
- **Site:** Dentigerous cysts may occur in association with any unerupted tooth, but most often involve **→Mandibular third molars**. Other frequent sites **→** include maxillary canines, maxillary third molars, and mandibular second premolars.
- **Signs & Symptoms:**
 - ✓ **Small dentigerous cysts** are usually asymptomatic and are discovered only on a routine radiographic examination, or when films are taken to determine the reason for the failure of a tooth to erupt.
 - ✓ **Large cysts** may be associated with a painless expansion of the bone in the involved area, *extensive* lesions may result in facial asymmetry.



- ✓ Dentigerous cysts may become infected and be associated with pain and swelling (Such infections is associated with a partially erupted tooth or by extension from a periapical or periodontal lesion that affects an adjacent tooth).

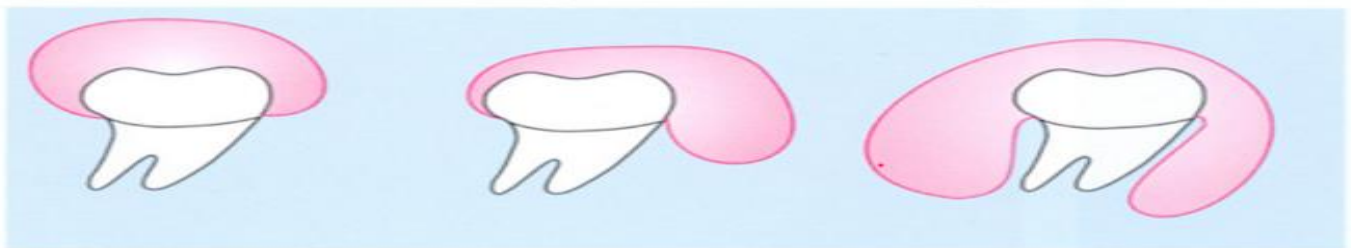
❖ **Radiographic Features:**

- ✓ Dentigerous cyst typically shows a unilocular radiolucent area that is associated with the crown of an unerupted tooth.
- ✓ The radiolucency usually has a well-defined and often sclerotic border, but an infected cyst may show ill-defined borders.
- ✓ Dentigerous cysts may displace the involved tooth for a considerable distance.
- ✓ Root resorption of adjacent erupted teeth can occur.

The cyst-to-crown relationship shows several radiographic variations:

1. **Central variety:** is the most common, the cyst surrounds the crown of the tooth and the crown projects into the cyst.
2. **Lateral variety:** is usually associated with mesioangular impacted mandibular third molars that are partially erupted. The cyst grows laterally along the root surface and partially surrounds the crown.
3. **Circumferential variant:** the cyst surrounds the crown and extends for some distance along the root so that a significant portion of the root appears to lie within the cyst.

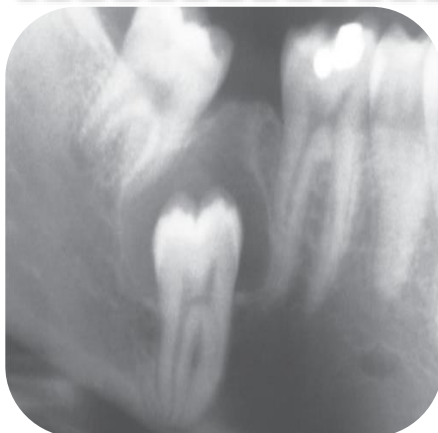
Dentigerous Cyst



Central

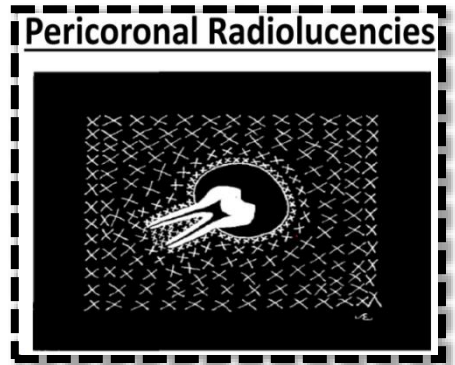
Lateral

Circumferential



✚ Differential Diagnosis of Peri-Coronal Radiolucency:

1. Dentigerous cyst
2. Unicystic ameloblastoma
3. OKC (in relation to a tooth)
4. Ameloblastic fibroma
5. Adenomatoid odontogenic tumor (AOT)
6. Hyperplastic follicle



❖ Histopathologic Features:

- Dentigerous cyst are lined by thin uniform layer (2-10 cell layers) of non-keratinized stratified squamous epithelium without rete ridges resembles the reduced enamel epithelium from which it is derived
- Mucous cells may be present → due to **metaplasia**.
- The lining is supported by a fibrous connective tissue capsule **free from inflammatory cells** unless gets secondary infected.
- Islands of odontogenic epithelium may be seen.
- **The cyst contains** a protein aqueous, yellowish fluid and cholesterol crystals. The soluble protein is around 5-7g/dl.

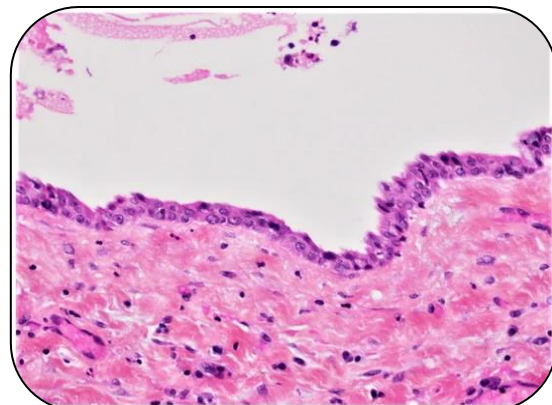
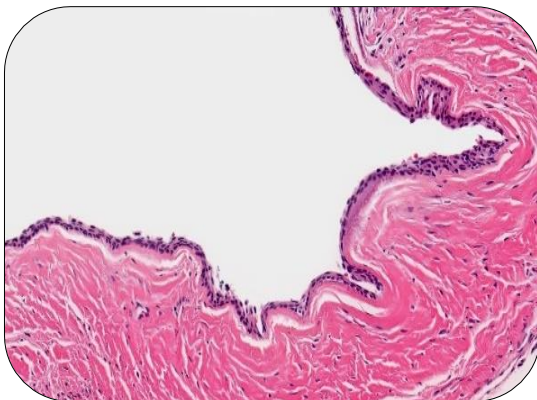
❖ Treatment and Prognosis:

- Enucleation of the cyst together with removal of the unerupted tooth.
- The prognosis is excellent, recurrence seldom is noted after complete cyst removal.

Complications of Dentigerous Cyst

The lining of a dentigerous cyst if not treated may undergo **neoplastic transformation** to:

1. Ameloblastoma.
2. Squamous cell carcinoma.
3. Intraosseous mucoepidermoid carcinomas may develop from mucous cells in the lining of a dentigerous cyst.



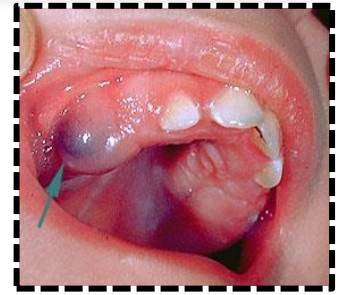
2-ERUPTION CYST (Eruption hematoma)

❖ Definition:

Eruption cyst is the soft tissue analogue of dentigerous cyst.

❖ Etiology & Pathogenesis:

- Accumulation of fluid or hemorrhage between crown of **erupting tooth** and the reduced enamel epithelium of the surrounding dental follicle, possibly as a result of trauma.
- Factors actually impeding eruption through soft tissues are **unknown**, possibly dense fibrous tissue.



❖ Clinical Features:

- The eruption cyst appears as a soft, often translucent swelling in the gingival mucosa overlying the crown of an erupting deciduous or permanent tooth.
- Most examples are seen in children younger than age 10.
- The cyst may occur with any erupting tooth but is *most commonly associated with the first permanent molars and the maxillary incisors*.
- Surface trauma may result in a considerable amount of blood in the cystic fluid, which imparts a blue color. Such lesions are referred to as eruption hematomas.

❖ Histopathological features:

- The epithelial lining of eruption cyst is similar to that of the dentigerous cyst (non-keratinized stratified squamous epithelium), so the eruption cyst is considered a superficial dentigerous cyst.
- The fibrous capsule shows inflammatory cells possibly as a result of trauma.
- The cystic cavity may contain blood in addition to the yellowish protein fluid as a result of trauma.

❖ Treatment & prognosis:

- It may **not be required** because the cyst usually ruptures spontaneously, permitting the tooth to erupt.
- If this does not occur, simple excision of the roof of the cyst generally permits speedy eruption of the tooth.

3-ODONTOGENIC KERATOCYST (OKC)

❖ Definition:

It is a developmental odontogenic cyst of with a characteristic lining of parakeratinized stratified squamous epithelium and is a distinctive form of odontogenic cysts due to:

- ➡ It may exhibit aggressive clinical behavior
- ➡ It has a significant recurrence rate.
- ➡ It may be associated with nevoid Basal Cell Carcinoma Syndrome (NBCCS).

❖ Origin:

- The cyst arises from **the remnants of the dental lamina**. Since the dental lamina gives rise to tooth primordium therefore this cyst was designated previously as **Primordial Cyst**.
- Origin of this cyst from the extension of basal cells of the overlying oral epithelium has also been suggested.

Growth of Odontogenic Keratocyst

- The odontogenic keratocyst has a destructive pattern of growth, burrowing through cancellous bone in a predominantly **anteroposterior direction**, this distinguishes it from the other odontogenic cysts that tend to expand in a **unicentric ballooning pattern**.
- The unique pattern of growth of the odontogenic keratocyst, in comparison to the other cysts, **suggests that different mechanisms of enlargement are involved:**
 - a. Active epithelial growth due to a higher rate of mitotic activity.
 - b. Cellular activity in the connective tissue capsules due to stimulation of osteoclasts.
 - c. Production of bone-resorbing factors including prostaglandins, collagenase, Interlukin1 and Interlukin 6.



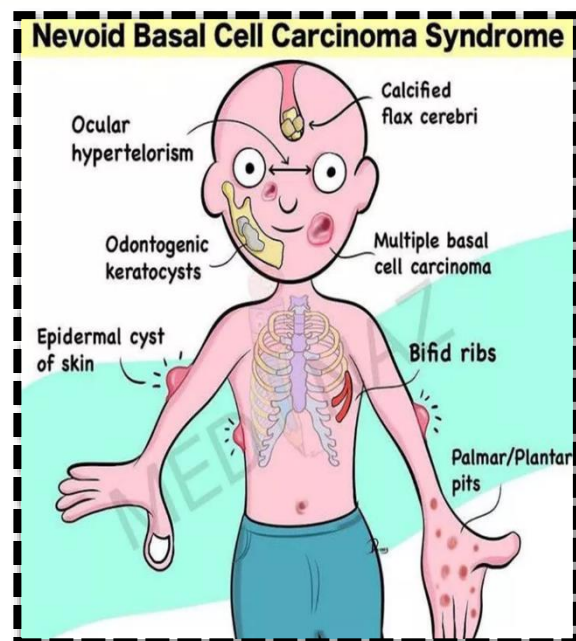
❖ Clinical features:

- **Incidence & Frequency:** OKC makes up 3% to 11% of all odontogenic cysts.
- **Age:** about 60% of all cases are diagnosed in people between **10 and 40 years of age**.
- **Sex:** There is a slight **male** predilection.
- **Site:** The **mandible** is involved in 60% to 80% of cases (marked tendency to involve the **posterior body and ascending ramus**).
- **Signs & Symptoms:**
 - ✓ **Small OKCs** ➡ are usually **asymptomatic** and discovered only during a radiographic examination.

- ✓ **Larger OKCs** → may be associated with pain, swelling, or drainage.
- ✓ **Some extremely large cysts** → may cause no symptoms. **Why???**
Because (OKCs tend to grow in an *anteroposterior direction* within the medullary cavity of the bone without causing obvious bone expansion. This feature may be useful in differential clinical and radiographic diagnosis because dentigerous and radicular cysts of comparable size are usually associated with bony expansion).
- It usually arises sporadically and presents as solitary lesions, but **multiple odontogenic keratocysts** may be developed and associated with **Nevoid Basal Cell Carcinoma Syndrome (NBCCS)**.

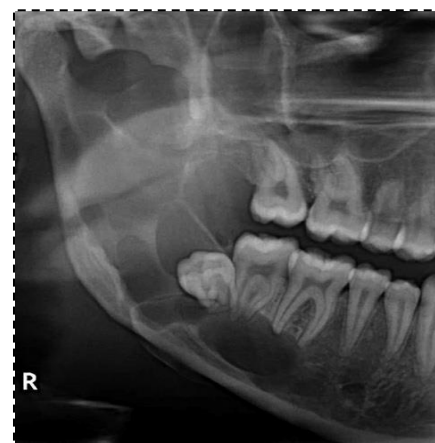
Nevoid Basal Cell Carcinoma Syndrome (NBCCS), (Gorlin-Goltz Syndrome)

- NBCCS is caused by mutation of a tumor suppressor gene in chromosome 9q t which affects signaling that is responsible for the growth of numerous tissues.
- **NBCCS has numerous manifestations including:**
 1. Multiple naevoid basal cell carcinoma occurs anywhere in the skin, especially on the face trunk, and neck it is seen in about 90% of cases.
 2. Multiple odontogenic keratocysts seen in about 90% of cases
 3. Skeletal abnormalities in ribs (bifid ribs), vertebral deformities (kyphoscoliosis), polydactyly, temporal & frontal posing.
 4. Central Nervous System disorders including calcified falx cerebri and brain tumours.
 5. Mild ocular hypertelorism.



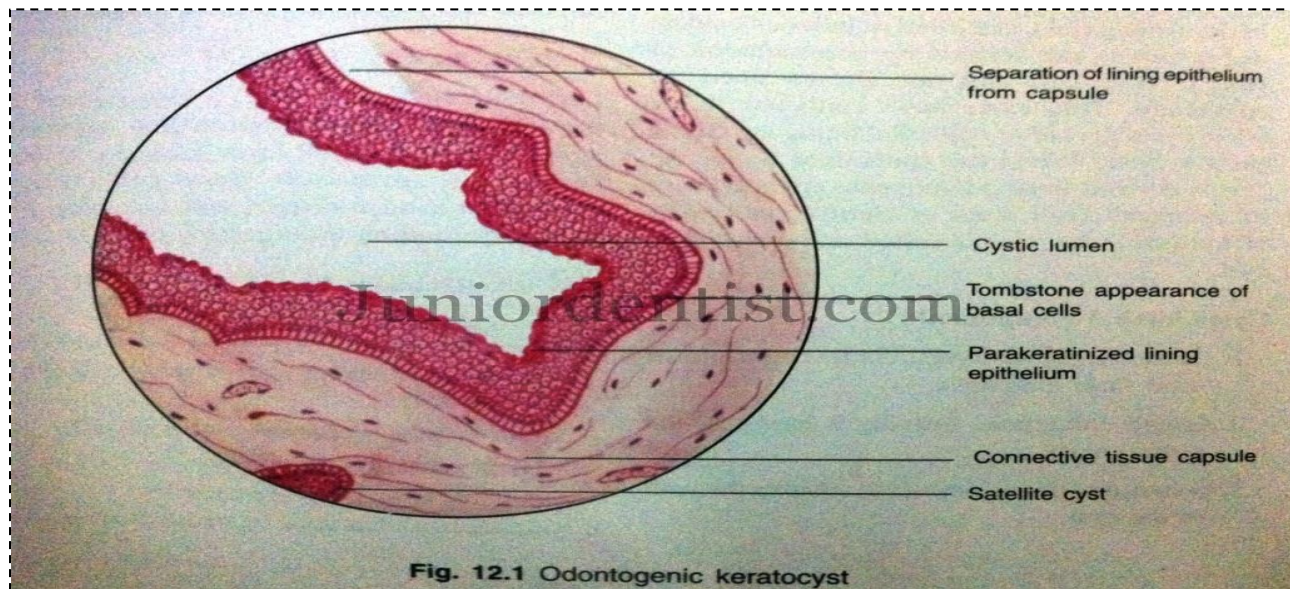
❖ **Radiographic Features:**

- OKC may be **Unilocular or multilocular radiolucency** which characteristically presents as a well-circumscribed radiolucency with smooth radiopaque margins.
- **Large lesions**, particularly in the posterior body and ascending ramus of the mandible, may appear multilocular.
- **An unerupted tooth** is involved in the lesion in 25% to 40% of cases.
- Resorption of the roots of erupted teeth adjacent to OKCs is less common than that noted with dentigerous and radicular cysts.



❖ Histopathologic Features:

- Odontogenic keratocysts contain thick, grey/white cheesy material consisting of keratin debris with little fluid and a low soluble protein level
- The cyst is lined by a regular layer of **parakeratinized stratified squamous** epithelium of some 5-10 cells in thickness.
- The epithelial lining has a well-defined basal epithelial cell layer is columnar or cuboidal cells with hyperchromatic, **palisaded arrangement** which is an important feature in distinguishing OKC from other cysts .
- The fibrous tissue capsule is usually thin and **free from inflammatory cells** and may contain satellite cysts called **(daughter cysts)**.
- The epithelium and connective tissue interface are usually flat, with **no rete ridge formation**, so, detachment of portions of the cyst-lining epithelium from the fibrous wall is commonly observed.
- The luminal surface shows flattened parakeratotic epithelial cells, which exhibit a wavy or **corrugated appearance**.
- **Small satellite cysts or daughter cysts**, cords, or islands of the odontogenic epithelium may be seen within the fibrous wall, these structures have been present in 7% to 26% of cases in various reported series.



❖ Treatment and Prognosis:

- Surgical excision with peripheral osseous curettage or ostectomy is the preferred method of management.
- It has a high recurrence rate after being removed.
- Follow-up examinations are important for patients with this lesion. Patients should be evaluated for completeness of excision, new keratocysts, and NBCCS.

Recurrence of Odontogenic Keratocyst

- In contrast to other odontogenic cysts, OKCs often tend to recur after treatment 30% or higher recurrence rate associated with OKCs due to:
 - 1) Thin epithelial lining is difficult to enucleate intact.
 - 2) Weak epithelial -connective tissue attachment, this causes the detachment of the epithelium and therefore more difficult to removal during surgery.
 - 3) A "new" cyst that has developed from **satellite cyst or daughter cyst** may be left behind after surgery
 - 4) Scalloped margins in multilocular variety.
 - 5) Extensive lesion involving coronoid/ascending ramus (difficult access).

❖ Differential Diagnosis:

- ✚ **In case of unilocular radiolucency-** Dentigerous cyst, COC, AOT, Unicystic ameloblastoma.
- ✚ **In case of multilocular radiolucency-** Conventional ameloblastoma, Central giant cell tumor, Aneurysmal bone cyst.

4-ORTHOKERATINIZED ODONTOGENIC CYST (OOC)

❖ Definition:

- It is a developmental odontogenic cyst that is predominantly lined by Orthokeratinized stratified squamous epithelium.
- It was originally referred to as a type of OKC but both clinically and histopathological differs from parakeratinized odontogenic cyst so they should be placed into a separated category
- Orthokeratinized odontogenic cyst are **not associated** with NBCCS., **do not have** high recurrence rates and **do not show** aggressive clinical behavior.
- Prominent palisaded basal layer of epithelial lining **is not** present in orthokeratinized odontogenic cyst which is diagnostic features in parakeratinized OKC.

▪ Etiology & Pathogenesis:

- The exact etiology and pathogenesis of OOC are not yet fully understood.
- It is believed to arise from the remnants of the dental lamina or the cell rests of Serres, which are remnants of the enamel organ.

❖ Clinical Features:

- **Age:** OOC most commonly affects individuals in the second to fifth decades of life.
- **Sex:** There is a slight predilection for males.
- **Site:** OOC often manifests in the posterior mandible, particularly in the molar and premolar regions. However, it can also occur in the maxilla and other regions of the jaws.

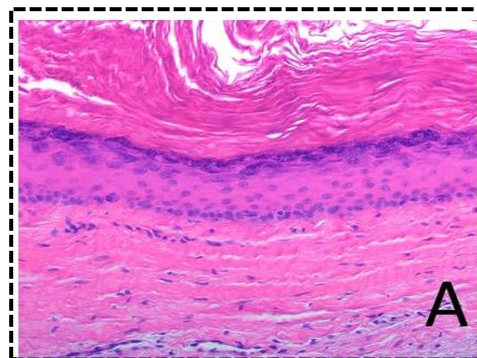
- **Symptoms:** OOC is typically asymptomatic and is often discovered incidentally during routine radiographic examinations. In some cases, patients may present with swelling, pain, or paresthesia if the cyst reaches a large size or impinges on adjacent structures.

❖ Radiographic Features:

- OOC: typically, present as well-defined unilocular radiolucent lesions. The borders of the cysts are usually corticated, indicating a slow and non-aggressive growth pattern.
- Most of OOC about 2/3 of cases are **associated with unerupted teeth**.
- Root Resorption: In some cases, OOCs may cause root resorption of adjacent teeth

❖ Histopathological Features:

- The hallmark histopathological feature of OOC is the presence of *orthokeratinized stratified squamous epithelium*, which is characterized by the absence of a granular layer.
- The granular cell layer is well defined and prominent which is absence in parakeratinized odontogenic cyst.
- The basal cells are cuboidal or flattened and absence palisaded basal cell layer



❖ Treatment & Prognosis:

- The treatment of OOC typically involves enucleation or marsupialization, depending on the size and location of the lesion.
- It has a lower recurrence rate than OKC.

The differences between OOC & OKC

Characteristic	OOC	OKC
Epithelial lining	Orthokeratinized stratified squamous epithelium	Parakeratinized stratified squamous epithelium
Granular layer	Absent	Present
Clinical presentation	Asymptomatic	Asymptomatic or associated with symptoms
Site of involvement	It can occur in any region of the jaws	Predilection for posterior mandible
Growth pattern	Slow and non-aggressive	Aggressive and infiltrative
Recurrence rate	Lower recurrence rate	Higher recurrence rate
Association with Gorlin syndrome	Not commonly associated	Frequently associated
Radiographic appearance	Well-defined unilocular radiolucency	Variable, can be unilocular or multilocular
Root resorption	Less commonly associated	More commonly associated
Treatment approach	Enucleation or marsupialization	Enucleation with adjunctive treatments
Prognosis	Generally favorable	Potentially more aggressive

5- LATERAL PERIODONTAL CYST (BOTRYOID ODONTOGENIC)

❖ Definition:

- It is an uncommon type of developmental odontogenic cyst (accounts for less than 2% of all epithelium-lined jaw cysts).
- It typically occurs along the lateral root surface of **a vital tooth**.

❖ Origin:

- It arises from the rests of the dental lamina
- It represents the intrabony counterpart of the gingival cyst of the adult.

❖ Clinical Features:

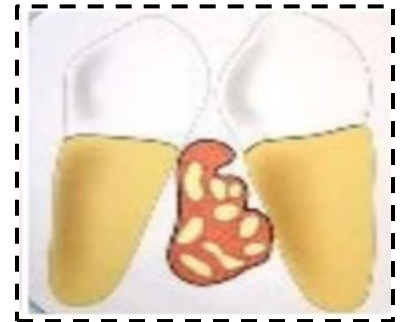
- **Age:** It most frequently occurs in patients in the 5th- 7th decades of life
- **Site:** 75% - 80% of cases occur in **the mandibular premolar–canine-lateral incisor area**.
- **Signs & Symptoms:** It is most often an asymptomatic lesion that is detected during a radiographic examination.

❖ Radiographic Features:

- The cyst appears as a well-circumscribed radiolucent area located laterally to the root of *the vital tooth*; most cysts are less than 1.0 cm in diameter.

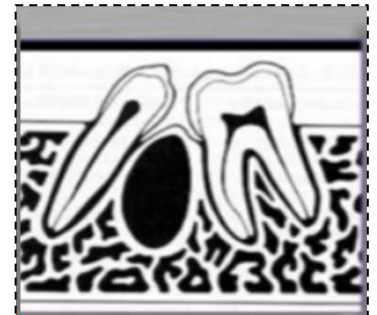
A variant of the lateral periodontal cyst (Botryoid Odontogenic Cyst)

- ✓ It has a polycystic appearance (botryoid odontogenic cysts).
- ✓ They grossly and microscopically show a grape-like cluster of small individual cysts.
- ✓ It is possibly the result of cystic degeneration and subsequent fusion of adjacent foci of dental lamina rests.
- ✓ The botryoid variant often shows a *multilocular radiographic appearance*.



✚ Differential Diagnosis of lateral Radiolucency:

1. Lateral Periodontal Cyst.
2. OKC that develops between the roots of adjacent teeth.
3. lateral radicular cyst.



❖ Histopathologic Features:

- The lateral periodontal cyst has a thin, generally non-inflamed, fibrous wall with an epithelial lining that is only 1 to 3 cells thick in most areas.
- This epithelium usually consists of flattened squamous cells, but sometimes the cells are cuboidal in shape.
- Some cysts show focal nodular thickenings of the lining epithelium, which is composed of glycogen-rich clear cells.
- Clear cell epithelial rests (represents rests of the dental lamina) are sometimes seen within the fibrous wall.

❖ Treatment and Prognosis:

- Conservative enucleation of the lateral periodontal cyst is the treatment of choice. Usually, this can be accomplished without damage to the adjacent teeth.
- Recurrence is unusual.

6-GINGIVAL CYST OF THE ADULT

❖ Definition:

The gingival cyst of the adult is an uncommon lesion; it is considered to represent *the soft tissue counterpart of the lateral periodontal cyst*, derived from rests of the dental lamina.

- It occurs in the free gingiva, attached gingiva, or sometimes in the interdental papilla

❖ Clinical Features:

- **Age:** Most commonly found in patients in the 5th and 6th decades of life.

- **Site:**

- ✓ Like the lateral periodontal cyst, it shows a striking predilection to occur in *the mandibular canine and premolar area* (60% - 75% of cases).
- ✓ They are almost invariably located on *the facial gingiva or alveolar mucosa*.
- ✓ Maxillary gingival cysts are usually found in the incisor, canine, and premolar areas.



- **Signs & Symptoms:**

- ✓ The cysts appear as painless domelike swellings, usually less than 0.5cm in diameter, often bluish or bluish-gray.
- ✓ In some instances, the cyst may cause a superficial "cupping out" of the alveolar bone. This is usually not detected on a radiograph but is apparent when the cyst is excised.

❖ Histopathologic Features:

Similar to those of the lateral periodontal cyst.

❖ Treatment and Prognosis:

Responds well to simple surgical excision; prognosis is excellent.

7- GINGIVAL (ALVEOLAR) CYST OF THE NEWBORN

❖ Definition:

- Gingival cysts of the newborn are small superficial keratin-filled cysts that are found on the alveolar mucosa of infants.

❖ Origin: These cysts arise from remnants of the dental lamina.

❖ Appearance:

- They are common lesions; however, because they disappear spontaneously by rupture into the oral cavity, the lesions are seldom sampled for biopsy.

✚ **N.B :** { similar inclusion cysts (e.g. Epstein's pearls and Bohn's nodules) are also found in the midline of the palate or laterally on the hard and soft palate }.

❖ Clinical Features:

- Gingival cysts of the newborn appear as small, multiple whitish papules on the mucosa overlying the alveolar processes of neonates (individual cysts no more than 2- 3 mm in diameter).
- The maxillary alveolus is more commonly involved than the mandibular.

❖ Histopathological Features:

- These are true cysts lined by stratified squamous epithelium and show a lumen filled with desquamated keratin, occasionally containing inflammatory cells

❖ Treatment and Prognosis:

No treatment is indicated because the lesions spontaneously involute after rupture; they are rarely seen after 3 months of age.

8- CALCIFYING ODONTOGENIC CYST (COC) OR GROLIN CYST

❖ Definition:

- ✓ It is a rare developmental odontogenic lesion characterized histologically by ghost cells.
- ✓ COC exists either as a cystic or a solid variant; the solid variant is known as Dentinogenic ghost cell tumor.
- ✓ The calcifying odontogenic cyst may **be associated with other recognized odontogenic tumors**, most commonly odontomas. However, adenomatoid odontogenic tumors and ameloblastoma also have been associated with calcifying odontogenic cysts.

❖ Origin:

Can arise from the dental lamina or remnant.

❖ Clinical Features:

- **Age:** 20-30 years.

- **Sex:** No sex predilection.
- **Site:** Most commonly in the maxilla.

Signs & Symptom

- Asymptomatic, slowly growing, and painless.

❖ **Radiographic Features:**

- o Well-defined unilocular or multilocular radiolucent area contains calcified masses or radiopaque masses.



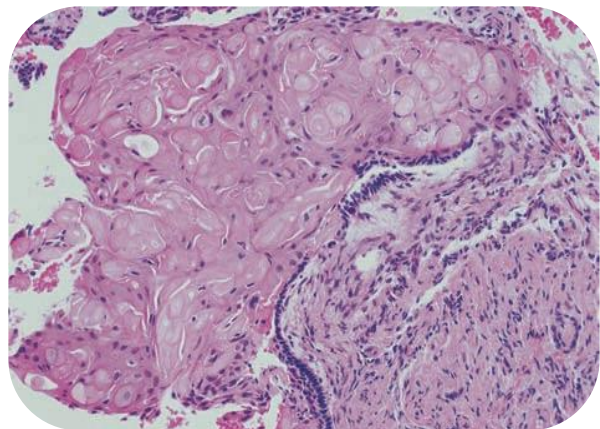
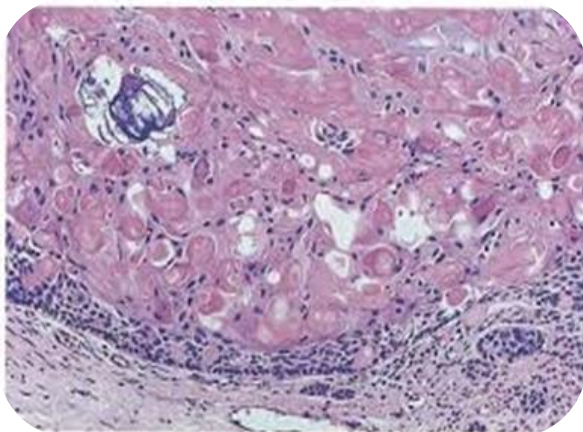
❖ **Histopathology features:**

The lesion is characterized by cystic spaces lined by epithelium, which shows:

- Basal layer of columnar cells (Ameloblast-like cells with reversed polarity) and overlying layers of loosely arranged cells (stellate reticulum-like cells).
- The most prominent and **unique microscopic feature** is the presence of **ghost cells**.
- **Ghost cells** are pale eosinophilic swollen cells that have lost their nucleus and nuclear membrane, and these cells undergo calcification.
- **Calcification** is seen as dentinoid, i.e., (dentine like irregular eosinophilic masses) or dystrophic calcification.
- The nature of the ghost cell is controversial. Some believe that this change represents coagulative necrosis or accumulation of enamel protein; others contend it is a form of normal or aberrant keratinization of odontogenic epithelium.
- All these structures can invade the cyst lumen and connective tissues, which elicits foreign body giant cells in C.T.

❖ **Differential Diagnosis:**

- 1- Odontogenic keratocyst.
- 2- Ameloblastoma.
- 3- Adenomatoid odontogenic tumor and ameloblastic fibro-odontoma



9- GLANDULAR ODONTOGENIC CYST (SIALO-ODONTOGENIC CYST)

❖ Definition:

- A glandular odontogenic cyst is a rare and **recently recognized** type of developmental odontogenic cyst that can show *locally aggressive recurrence potential*.
- It has some histologic features that suggest a mucus-producing salivary gland tumor (**low-grade mucoepidermoid carcinoma**).

❖ Origin:

Although it is accepted as being of odontogenic origin, it also shows glandular or salivary features that presumably are an indication of the pluri-potentiality of odontogenic epithelium.

❖ Clinical Features:

- **Age:** The glandular odontogenic cyst occurs most commonly in middle-aged adults with a mean age of 49 years; rarely does it occur **before** the age of 20.
- **Site:** Nearly 85% of cases occur in the mandible. The cyst has a strong predilection for *the anterior region of the jaws*, and many mandibular lesions will cross the midline.



Signs & Symptoms:

- The size of the cyst can vary from small lesions less than 1cm in diameter to large destructive lesions that may involve most of the jaw.
- Small cysts may be asymptomatic; however, large cysts often produce clinical expansion, which sometimes can be *associated with pain or paresthesia*.

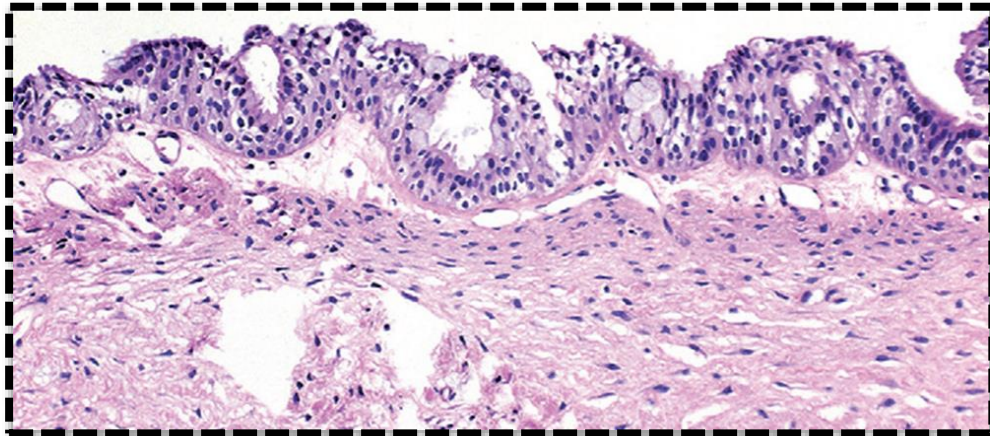
❖ Radiographic Features:

- Unilocular or, more commonly *a multilocular radiolucency*. The margins of the radiolucency are usually well-defined with a sclerotic rim.
- More aggressive lesions have shown an ill-defined peripheral border.

❖ Histopathologic Features:

- The cyst is lined by squamous epithelium of varying thickness.
- The interface between the epithelium and the connective tissue wall is generally flat.
- The superficial epithelial cells that line the cyst cavity tend to be cuboidal to columnar and have an irregular and sometimes papillary surface. Occasionally cilia may be noted.
- **Pools of mucinous material** are often present within the epithelium. Cuboidal cells usually line these pools.
- **Glandular duct-like spaces** within the epithelial lining are another characteristic finding
- Mucous cells may or may not be present within the epithelium.

- The fibrous wall is usually devoid of any inflammatory cell infiltrate.
- ❖ **Treatment and Prognosis:**
- Most cases of glandular odontogenic cyst have been treated by enucleation or curettage.
- The cyst shows a tendency for recurrence (30% or more of all cases).



Glandular odontogenic cyst lined by epithelium showing ductlike features and mucous cells.

Non-Odontogenic Cysts

➤ **Non-odontogenic cysts :**

Cysts that arise from epithelial inclusion or entrapment in the lines of fusion of the developing facial processes during embryogenesis.

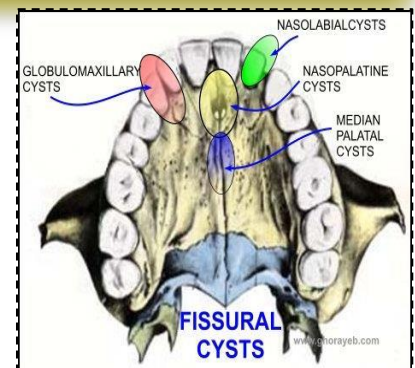
1-NASOPALATINE DUCT (INCISIVE CANAL) CYST

❖ **Definition:**

- The Nasopalatine Duct Cyst (also known as Incisive Canal Cyst) is **the most common** non-odontogenic developmental cyst that arises from remnants of the nasopalatine duct.

❖ **Etiology & Pathogenesis:**

- It arises from epithelial remnants of the nasopalatine duct (which connects the oral and nasal cavities in the embryo) so it is classified as a “fissural cyst”.
- It may arise at any point along the nasopalatine canal; most cases arise in the lower part of the canal, but some may arise within *palatal soft tissues* at the point of opening of the canal, where the lesions are called ➡ ***Cysts of the palatine papilla or Cyst of incisive papilla.***



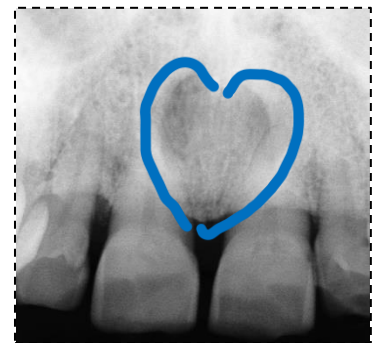
- The stimulus for cyst formation is unknown but:
 - ***The bacterial infection and/or trauma*** is thought to have a role.
 - The mucous glands within the lining may cause cyst formation as a result of mucin secretion.

❖ Clinical Features:

- **Frequency & Incidence:** commonest non-odontogenic developmental cyst (occurring in about 1% of the population).
- **Age:** 5th & 6th decades.
- **Sex:** males more than females.
- **Site:** Found in the midline of the anterior maxilla, just behind the central incisors. It may be seen in the soft tissue of the incisive papilla or as a cyst of papilla palatine
- **Signs & Symptoms:**
 - ✓ Asymptomatic (unless infected) & discovered on routine x-ray examination.
 - ✓ It may present as a slowly growing swelling in the anterior region of the midline of the palate and may discharge in the mouth through a sinus giving (a salty taste) and **pain (pressure on nerves).**

❖ Radiographic Features:

- A well-defined round, ovoid, or **heart-shaped radiolucency** with a sclerotic rim.
- Usually symmetrical about the midline between the roots of the central incisors, but some are displaced to one side.
- The classic heart shape as a result of *superimposition of the anterior nasal spine*.
- Its radiographic diameter can range from (6mm) to (6 cm).



✚ Differential Diagnosis Radiographically:

1. Normal incisive fossa (not greater than 6 mm wide).
2. Radicular cyst of incisors (non-vital teeth).

❖ Histological Features:

▪ **The epithelial lining:**

The cysts may be lined by a variety of different types of epitheliums:

- a.** Stratified squamous epithelium, **b.** pseudostratified ciliated columnar (respiratory) epithelium often containing mucous cells, **c.** cuboidal epithelium, may be seen alone or in an combination → *This is according to relation to the nasal or the oral cavities.*

▪ **Connective tissue wall:**

- ✓ Prominent neurovascular bundles from the terminal branches of the long sphenopalatine nerve and vessels helpful diagnostic aid.
- ✓ Mucous glands & chronic inflammatory cell infiltration.

❖ Treatment and Prognosis:

Treated by surgical enucleation, Recurrence is rare.

2-NASOLIBAL CYST

Nasolabial Cyst

❖ Definition:

- It is a rare lesion which arises in the soft tissue of the upper lip lateral to midline, just below the ala of the nose.
- It is traditionally regarded as a jaw cyst, although strictly speaking it should be *classified* ➡ *as a soft tissue cyst*.

❖ Etiology & Pathogenesis:

Its pathogenesis is uncertain, although there are two major theories:

- **One theory** ➡ considers the nasolabial cyst to be a “fissural cyst” arising from epithelial remnants entrapped along the line of fusion of maxillary, medial nasal, and lateral nasal processes.
- **A second theory** ➡ suggests developing from remnants of the lower part of the embryonic nasolacrimal duct.

❖ Clinical Features:

- **Frequency:** rare in occurrence.
- **Age:** most commonly seen in adults (peak prevalence in 4th -5th decades of life).
- **Sex:** a significant predilection for women (F: M ratio = 3: 1).
- **Signs & Symptoms:** slowly growing swelling & occasionally, pain and difficulty in nasal breathing.
 - o Extra orally ➡ obliterating the nasolabial fold & distorting the nostril.
 - o Intra orally ➡ bulge in the labial sulcus.
 - o Fluctuant lesion & may arise bilaterally.



❖ Radiographic Features:

- Difficult to interpret on radiograph; **negative on X-ray**.
- May be seen as localized increased lucency of alveolar process above apices of incisors.

❖ Histological Features:

- The cysts are usually lined by pseudostratified columnar epithelium but stratified squamous epithelium, mucous cells, and ciliated cells may also be present.
- Supported by CT capsule.
-

❖ Treatment & Prognosis: complete surgical excision of cyst.

3-POST SURGICAL CILITED CYST

Nasolabial Cyst

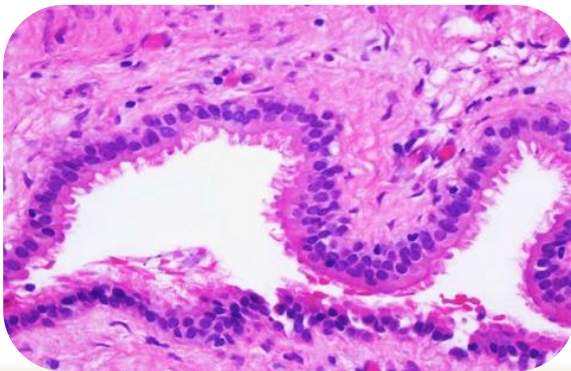
❖ **Definition:**

- ✓ It is also known as a postoperative maxillary cyst—a rare, non-odontogenic cyst lined by respiratory epithelium, with a fluid-filled cavity that can develop years or even decades after a surgical procedure or trauma involving the upper jaw or sinuses

❖ **Clinical Features:**

- Localized swelling in the cheek or jaw area, **mild pain**, or a feeling of pressure.
- Asymptomatic: Many cases cause no pain at all and are only discovered by accident on routine dental X-rays or CT scans.
- It occurs most commonly in the posterior maxilla but is very rare in the mandible.

- ❖ **Histopathologically:** the cyst is lined by pseudostratified ciliated columnar and mucous cells.



4-MEDIAN PALATAL CYSTS

❖ **Nasolabial Cyst**

- It is very rare and is thought to originate from epithelial remnants at the time of fusion of the palatal shelves of the maxilla.

❖ **Clinical Features:**

- Firm or fluctuant swelling in the midline of the hard palate posterior to the incisive canal.
- Asymptomatic; some complain of pain or expansion.

❖ **Radiographic Features:**

- Well-defined radiolucency is demonstrated in the midline of the hard palate, surrounded by a radiopaque margin and divergence of central incisors

❖ **Histopathological Features:**

- The histological features are similar to those of a nasopalatine duct cyst, and some published cases may represent displaced nasopalatine duct cysts.

- ❖ **Treatment:** Surgical removal

Non-epithelialized primary bone cysts

1-SIMPLE BONE CYST

(TRAUMATIC BONE CYST, HEMORRHAGIC BONE CYST, SOLITARY BONE CYST, IDIOPATHIC BONE CAVITY)

❖ Definition:

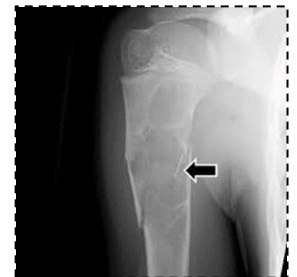
It is a benign, empty, or little fluid-containing cavity within bone that lacks an epithelial lining.

❖ Etiology & Pathogenesis: *The cause and pathogenesis are uncertain & controversial; several theories have been proposed:*

1. **The trauma-hemorrhage theory** (designated traumatic bone cyst), trauma to the bone that is insufficient to cause a fracture results in an intraosseous hematoma. If the hematoma does not undergo organization and repair, it may liquefy, resulting in a **cystic defect (an empty cavity)**.
2. **Other theories** include inadequate venous drainage, local disturbance in bone growth, ischemic marrow necrosis, and localized alteration in bone metabolism resulting in **osteolysis**.

❖ Clinical Features:

- **Age:** children & adolescents with a peak incidence in 2nd decade.
- **Sex:** about 60% of cases occur in males.
- **Site:**
 - Most commonly involve long bones, but occasionally seen in the jaws.
 - In the jaws are restricted to **the mandible premolar & molar areas**.
- **Signs & Symptoms:**
 - It is asymptomatic & usually found on routine radiographs examination.
 - Only 25% of cases may show expansion of bone.



❖ Radiographic Features:

- The lesion presents as a radiolucency of variable size (1 to 10 cm in diameter) & irregular outline.
- Interdental scalloping around & between the roots of standing teeth is seen.
- The margins of the lesion are usually well-defined.
- The involved teeth are vital and do not show root resorption.



❖ Histological Features:

- **Surgical exploration:** *to confirm diagnosis* reveals a rough bony-walled cavity with no soft tissue lining. It is empty or, in some cases, shows little clear or blood-stained fluid.
- **Histopathologic examination of curettage shows:**
 - ✓ Delicate layer of loose, vascular CT containing extravasated RBCs & deposits of hemosiderin pigment.
 - ✓ No epithelial lining.

❖ Treatment & Prognosis:

Rapid healing follows surgical exploration, but even without surgical intervention the cyst will resolve *spontaneously* with time. Prognosis is excellent.

2-ANEURYSMAL BONE CYST (ABC)

❖ Nasal Labial Cyst

It is an intra-osseous accumulation of variable-sized blood-filled spaces surrounded by cellular fibrous CT that often is admixed with trabeculae of reactive woven bone.

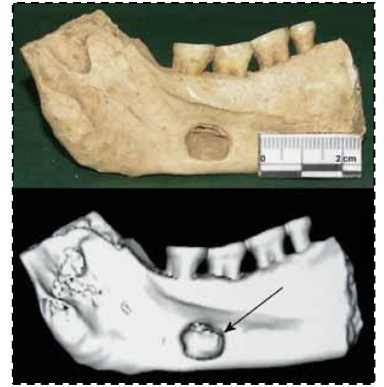
❖ Etiology & Pathogenesis:

- The cause & pathogenesis of the aneurysmal bone cyst are poorly understood.
- Several investigators have proposed that aneurysmal bone cyst arises from a traumatic event, vascular malformation, or neoplasm that disrupts the normal osseous haemodynamics and leads to an enlarging haemorrhagic extravasation.
- It occurs either as **a primary or as a secondary lesion** in a pre-existing intra-bony lesion) as  (fibro-osseous lesions & central giant cell granuloma).

❖ Clinical Features:

- **Age:** (most cases arise in children & young adults with a mean age of 20 years).
- **Sex:** No significant sex predilection is noted.
- **Site:**
 - It is located most commonly in the shaft of a long bone or in the vertebral
 - Gnathic aneurysmal bone cysts are uncommon (only 2% reported from the jaws).
 - In the jaws, mostly affects mandible posterior part of the body or the angle.
- **Signs & Symptoms:**
 - ✓ The most common clinical manifestation is a swelling that has usually **developed rapidly**; **Pain** is often reported.
 - ✓ The lesion is known to cause facial deformity, migration of teeth, malocclusion, cortical thinning, perforation, and pathologic fracture.
 - ✓ Maxillary ABCs may cause nasal obstruction, bleeding, and proptosis.

- ✓ **At the time of surgery**, when the periosteum and bony shell are removed, dark venous blood frequently wells up and venous-like bleeding may be encountered. The appearance at surgery has been likened to that of a **"blood-soaked sponge."**



❖ **Radiographic Features:**

- Usually shows a multilocular, soap bubble appearance.
- May also show a unilocular appearance.
- Displacement of teeth and root resorption.
- Cortical expansion, thinning and perforation.
- **"Ballooning" or "blow out"** expansion of bone (the cortex literally expands like a blown balloon).

❖ **Histopathologic Features:**

- Blood-filled spaces of varying size, surrounded by cellular fibroblastic tissue containing **multinucleated giant cells**, and trabeculae of osteoid and woven bone.
- The blood-filled spaces are not lined by endothelium.

❖ **Treatment and Prognosis:**

Curettage or enucleation, sometimes supplemented with cryosurgery.

✚ **Differential Diagnosis of Aneurysmal bone cyst:**

- Radiographically should be differentiated from ➡ Multilocular Radiolucent Lesions.
- Microscopically should be differentiated from ➡ Central Giant Cells Lesions.

**3-STAFNE, S BONE CYST
(LINGUAL MANDIBULAR SALIVARY GLAND DEPRESSION;
LATENT BONE CYST; STATIC BONE CYST)**

Nasolabial Cyst

❖ **Definition:**

This condition represents a focal concavity of the cortical bone on the lingual surface of the mandible, containing ectopic normal salivary gland tissue.

❖ **Clinical Features:**

- A striking male predilection is observed (80% to 90% of all cases seen in men).
- It is symptomless & found by chance in routine radiograph examinations.
- Stafne defects typically **remain stable in size**; hence the name static bone cyst.

❖ **Radiographic Features:**

- It appears as a round or oval, well-demarcated radiolucency between the premolar region & angle of the mandible.
- It is usually located beneath the inferior dental canal and is bilateral as a saucer-shaped depression on the lingual aspect of the mandible.

N.B.: The diagnosis can usually be made on a clinical basis by the typical radiographic location and lack of symptoms; clinical diagnosis can be confirmed by CT scans, MRI, or sialography.

❖ **Histopathologic Features:**

- Because of the typical radiographic appearance, biopsy is usually not necessary.
- If biopsy is performed, normal submandibular gland tissue is usually seen.

❖ **Treatment and Prognosis:**

No treatment is necessary for patients with Stafne defects of the posterior mandible, and the prognosis is excellent.